

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
BANGLADESH ARMY UNIVERSITY OF SCIENCE & TECHNOLOGY (BAUST)
SAIDPUR CANTONMENT, NILPHAMARI

(Project/Thesis Proposal)

Application for the approval of B.Sc. Engineering Project/Thesis
(Computer Science & Engineering)

Session: Winter 2023

Date: April 11, 2023

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4. Name of the Department : Computer Science & Engineering

Program : B.Sc. Engineering

**5. Date of First Enrolment
in the Program** : 29 December 2019

6. Tentative Title : Bengali Word Assimilation Integrating Using Deep Learning
Methods.

7. Introduction

One of the most extensively spoken languages in the world is Bengali. Over 98% of Bangladeshis plus a sizable percentage of Indians speak this language as their mother's tongue. A lot of individuals prefer digital paper as it can be viewed on laptops, mobile phones, and other gadgets used in the ever-expanding realm of social media. Because of this, there are numerous Bangla news websites that offer the most recent information on a variety of subjects. A number of news-based websites are also built on various online platforms like Facebook and Instagram. Every online Bangla news provider has a different layout, style, and categorization system to convey the news items. A lot of people who read newspapers enjoy reading and evaluating news from different sources. These varied portals generate content for almost every categorization; thus, it might not always be possible to classify these enormous collections of material into the appropriate categories. Online news websites include subject categories and subcategories, which vary significantly across newspapers. These might not be enough to pique readers' curiosity as a result. Because readers like to research news from several sources instead of just one, providing them with relevant news depending on its contents may improve their experience. The article's main objective is to help Bengali internet news readers find relevant material by using multi-category classification. Our main objective is to construct a computerized framework for the vital task of classifying Bangla reports based on their substance, which is important to readers. Over the past few decades, many methods from machine learning and statistics have been applied to extract meaningful features and efficiently classify textual data. These supervised learning techniques require a lot of textual data to be trained.

8. Background and Present State of the Problem

There is plenty of data and information about news and events in the modern world. Information is easily accessible to people nowadays thanks to modern communication technology. Data accessibility is not an issue, but it becomes risky when the data contains false information or stories that go viral globally. This fake news has an immoral impact on social and personal relationships and has the potential to cause misunderstandings and even political tension. According to a recent survey, roughly one in three people in the US, Spain, Germany, the UK, Argentina, and the Republic of Korea say they have come across false or misleading information about COVID-19 on social media [1]. The effects of fake news are wreaking havoc on the world. In Bangladesh, several pagodas were set on fire in 2012 after a picture appeared that depicted the Quran in an offensive way. The young Buddhist who was tagged in the picture was not in any way associated with the picture. Even though the identity of the image could not be confirmed, many rumors circulated as a result of it [2]. A false rumor about the Padma Bridge's officials giving up people's lives at the building site went viral online in 2019. This rumor then raises the possibility that strangers are abducting children [3]. An example of this can be found in Bangladesh in the

Ramu incident of 2012, when nearly 25,000 people joined forces to abolish Buddhist temples based on a post on Facebook from a fictitious account [4]. Thankfully, there are websites dedicated to combating fake news. However, these websites are unable to adequately address the majority of fake news incidents. Certain computational techniques are employed to separate false information, which is detrimental to our day-to-day existence. There is a lot of research being done on English languages these days. Currently, over 341 million people use Bangla to communicate worldwide. However, there is no tool to identify false information written in Bangla. Naturally occurring language processing tasks for regional languages are challenging because of these incredibly low resources. Regional language assets are therefore still needed for a variety of tasks related to natural language processing.

9. Objective with Specific Aims and Possible Outcome

Our work focuses on categorizing a dataset using a variety of deep learning and machine learning algorithms, such as CNN, GRU, BiLSTM, LSTM, and BERT. The primary aim is to assess their performance using metrics like precision, recall, F1 score, as well as through visual aids like paradigms and diagrams. This evaluation will help determine which algorithm yields the most proficient results. By comparing the outputs of these algorithms, you'll be able to discern which one performs best. This assessment isn't solely about the accuracy of the categorization but also involves evaluating how well these algorithms handle the training and testing data, shedding light on their adaptability and generalization capabilities. Our study design is comprehensive, encompassing a range of contemporary algorithms used in natural language processing (NLP). The utilization of various evaluation metrics will provide a nuanced understanding of their strengths and weaknesses in document categorization. This holistic approach allows for a more informed conclusion about which algorithm might be most suitable for this specific task.

10. Outline of Methodology Design

The work methodology involves several crucial steps. It commences with manual collection of the dataset from social media platforms like Facebook. Subsequently, data preprocessing takes precedence, recognizing its pivotal role in preparing data for effective deep learning models. Tokenization follows, splitting the dataset into training and testing subsets. The subsequent phase involves model creation, drawing upon five distinct deep learning models: CNN, GRU, BiLSTM, LSTM, and BERT [16]. Each model undergoes training, a fundamental phase to optimize its performance. The work culminates in performance analysis, evaluating the models based on various metrics. This structured methodology ensures a comprehensive approach—from data collection to model training and performance assessment—essential for deriving meaningful insights into the efficiency of each model in document categorization tasks.

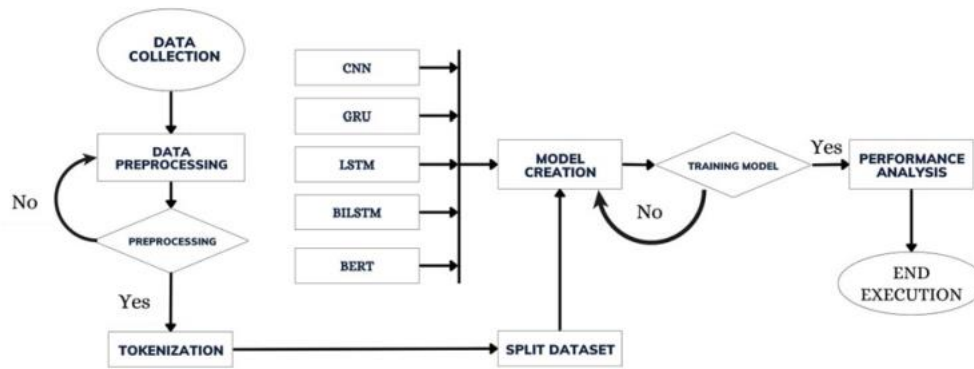


Figure 1: Flow Chart of Our Work.

11. Resources Required to Accomplish the Task

- Google Colaboratory
- Google Drive
- Deep learning tools
- Pandas
- Matplotlib

12. References

- [1] R. Nielsen, R. Fletcher, N. Newman, J. Brennen, and P. Howard, Navigating the ‘infodemic’: How people in six countries access and rate news and information about coronavirus. Reuters Institute for the Study of Journalism, 2020.
- [2] “Mobs beat five dead for ‘kidnapping’ — thedailystar.net,” <https://www.thedailystar.net/frontpage/news/mobs-beat-2-dead-kidnapping-1774471>.
- [3] M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar, “Banfakenews: A dataset for detecting fake news in bangla,” arXiv preprint arXiv:2004.08789, 2020.
- [4] T. Islam, S. Latif, and N. Ahmed, “Using social networks to detect malicious bangla text content,” in 2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT). IEEE, 2019, pp. 1–4.
- [5] A. Dhar, H. Mukherjee, N. S. Dash, and K. Roy, “Automatic categorization of web text documents using fuzzy inference rule,” *Sadhanā*, vol. 45, pp. 1–22, 2020.

[6] A. Dhar, N. S. Dash, and K. Roy, “Classification of bangla text documents based on inverse class frequency,” in 2018 3rd International Conference On Internet of Things: Smart Innovation and Usages (IoT-SIU). IEEE, 2018, pp. 1–6.

[7] Dhar, “Application of tf-idf feature for categorizing documents of online bangla web text corpus,” in Intelligent Engineering Informatics: Proceedings of the 6th International Conference on FICTA. Springer, 2018, pp. 51–59.

13. Cost Estimation

- a) **Cost of Materials:3000**
- b) **Cost of Repot Printing and Binding:3000**
- c) **Others: 12000 for final paper submission and published**

14. Committee for Advance Studies and Research (CASR)

Meeting No:

Resolution No:

Date:

15. Number of Under-Graduate Students Working with the Supervisor at Present: 12

Signature of the Students	Department of CSE
	----- Signature of the Supervisor
	----- Signature of the Head of the Department